THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

May 16, 2015 TIME: 9:30 am – 12:30 pm

I. Answer ALL questions. Section A carries 60 marks, and Section B carries 40 marks.

II. Atomic weights: H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K 39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

III. Properties of water:

Density = 1000 kg/$m^3$
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/kg.°C
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

**Section A. (20 multiple choice questions with 3 marks each)**

1. Carbon monoxide (CO) concentration at a major highway reaches 12 mg/m$^3$ at 25 °C. Estimate its concentration in ppmv. (a) 0.4 ppmv, (b) 1 ppmv, (c) 10 ppmv, (d) 20 ppmv, or (e) 250 ppmv.
2. Which of the following processes is not suitable for hardness removal? (a) Nanofiltration, (b) ion exchange, (c) lime-soda method, or (d) air stripping.
3. A radioactive chemical is degraded in a batch reactor with a first-order rate constant of 0.1 day$^{-1}$. If the initial mass of the chemical is 1 mg, how many days does it take to reduce its mass to 0.001 mg? (a) 1 day, (b) 3 days, (c) 6.9 days, (d) 10 days, (e) 69 days, or (f) 100 days.
4. For the chemical in Question #3, estimate its half-life. (a) 1 day, (b) 3 days, (c) 6.9 days, (d) 10 days, (e) 69 days, or (f) 100 days.
5. A groundwater has the following properties: pH 7.8, [$HCO_3^-$] = 60 mg/L, [$Mg^{2+}$] = 12 mg/L and [$Ca^{2+}$] = 40 mg/L. Determine its total hardness as $CaCO_3$(s). (a) 40 mg/L, (b) 50 mg/L, (c) 52 mg/L, (d) 60 mg/L, (e) 72 mg/L, (f) 99 mg/L, or (g) 150 mg/L.
6. Which type of membrane is most suitable for membrane bioreactors (MBR)? (a) Reverse osmosis membrane, (b) nanofiltration membrane, (c) microfiltration membrane, or (d) ion exchange membrane.
7. Which of the following biological processes is an attached-growth process? (a) Tricking filter, (b) aerated lagoon, (c) activated sludge process, (d) membrane bioreactor, or (e) oxidation pond.
8. Membrane filtration and UV disinfection are used in series for pathogen removal. The membrane filtration step has a 3-log removal of pathogen, and the UV step kills 99% of the pathogen. What is the overall efficiency of the combined processes? (a) 2-log, (b) 3-log, (c) 4-log, (d) 99.99%, (e) 99.999%, or (f) none of the above.
9. Which of the following reactions is not involved in a typical anaerobic digestion process? (a) Nitrification, (b) hydrolysis, (c) acidogenesis, or (d) methanogenesis.
10. A wastewater sample has an estimated $BOD_5$ of approximately 500 mg/L. Which of the following dilution factors is the most appropriate to dilute the wastewater sample for BOD test? (a) 1:2 dilution, (b) 1:10 dilution, (c) 1:100 dilution, (d) 1:1000 dilution, or (e) no dilution needed.
11. Thermal evaporation (distillation) can be used to produce freshwater from seawater. Estimate the amount of thermal energy required to distill 1 m$^3$ of water. Give your answer in kWh. Hint: 1 kWh = 3.6 MJ. (a) 0.7 kWh, (b) 70 kWh, (c) 110 kWh, or (d) 700 kWh.
12. A sedimentation tank has a removal efficiency of 100% for sand particles with a diameter of 100 μm. What is the removal efficiency for sand particles with a diameter of 10 μm? (a) 1%, (b) 10%, (c) 100%, (d) ≥ 1%, or (e) ≤ 10%.
13. Which of the following disinfectants has the strongest disinfection power? (a) Hypochlorite salt, (b) hypochlorous acid, (c) ozone, (d) chlorine dioxide, or (e) chloramine.
14. Which of the following mechanisms is not relevant to coagulation and flocculation processes? (a) Charge neutralization, (b) trapping, (c) inter-particle bridging, (d) oxidation, or (e) none of the above.
15. Solar energy reaches the Earth's surface by: (a) Conduction, (b) convection, (c) radiation, (d) conduction and convection, or (e) none of the above.
16. Treating every kg of food waste in an anaerobic digester produces approximately 0.5 m$^3$ of methane gas (at 25 °C). The higher heating value of methane is 890 kJ/mol. Estimate

the amount of bioenergy contained in the methane gas. (a) 18 kJ, (b) 36 kJ, (c) 445 kJ, (d) 18 MJ, (e) 36 MJ, or (f) 445 MJ.

17. $CaF_2$ is dissolved in distilled water. When equilibrium is reached, the calcium concentration in water is 0.2 mM. Determine the $pK_{so}$ of $CaF_2$. (a) 3.4, (b) 6.8, (c) 7.1, (d) 10.5 or (e) 11.1.
18. Which of the following effects is not directly related to increased atmospheric $CO_2$ concentration? (a) Sea level rise, (b) accelerated melting of glaciers, (c) ocean acidification, (d) more frequent extreme weather conditions, (e) increased UV exposure, or (f) potential change in ocean circulation pattern.
19. Which of the following air pollutants is a secondary pollutant? (a) CO, (b) ground-level ozone, (c) $NO_2$, (d) $SO_2$, or (e) none of the above.
20. What is the nitrogenous oxygen demand of a wastewater sample containing 100 mg/L $C_6H_{12}O_6$? (a) 80 mg/L, (b) 100 mg/L, (c) 120 mg/L, (d) 160 mg/L, or (e) none of the above.

**Section B (Two questions with 20 marks each)**

21. Ocean is considered as a major sink of the greenhouse gas carbon dioxide. The following information are also available:
    - Atmospheric $CO_2$ concentration: 370 ppm
    - Henry's constant for $CO_2$ at 25°C: $K_H = 3.39 \times 10^{-2}$ M/atm
    - Acidity constants of carbonic acid ($H_2CO_3$) at 25°C: $pK_1 = 6.35$ and $pK_2 = 10.33$
    - pH of ocean water: 8.1

    (a) Determine the molar concentration of $H_2CO_3$, $HCO_3^-$, and $CO_3^{2-}$. Which species is the dominant species?

    (11 marks)

    (b) Determine the alkalinity of the ocean water. Give your answer in meq/L.

    (3 marks)

    (c) Briefly discuss how you can estimate the total amount of inorganic carbon stored in the ocean. What additional information is needed? Also list the major assumptions required. You do not need to show any numerical calculations.

    (6 marks)

22. (a) Meteorology plays a very important role in air pollution.
    (i) With the aid of simple sketches, briefly explain the concepts of neutrally-stable atmosphere, unstable atmosphere, and stable atmosphere.

    (12 marks)

    (ii) What is temperature inversion and how does it affect air pollution?

    (3 marks)

    (b) A wastewater has a $BOD_5$ of 200 mg/L at 20°C. The ultimate BOD is 280 mg/L. Determine the rate constant at 20°C. What is the $BOD_5$ at 25°C? Assume a temperature correction factor of 1.047 in your calculation.

    (5 marks)

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